



Kolegji AAB
CILËSI. LIDERSHIP. SUKSESI

CLOUD COMPUTING AND MOBILE CLOUD

FAKULTETI I SHKENCAVE KOMPJUTERIKE

LËNDA: CLOUD COMPUTING AND MOBILE CLOUD – ECTS 6

DREJTIMI: PROGRAMIM BA PROFESONALE

VITI III - SEMESTRI VI

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06.03. & 12.03.2026

Cloud computing and mobile cloud

- Ligjeratat: të enjteve | 10:45 - 12:15 | A-202
- Ushtrimet: të enjteve | 09:30 - 10:30 | A-Lab II
- Ushtrimet: të enjteve | 12:30 - 14:00 | A-Lab II
- Konsultimet para/pas ligjerateve/online
- Terminet përcaktohen gjatë orëve mësimore

Literatura

- **Literatura themelore**
- Bhowmik, S. (2018). Cloud Computing. Cambridge University Press.
- De, D. et al. (eds.) (2022). Green Mobile Cloud Computing. Springer.
- Amazon Web Services. (2024). Overview of Amazon Web Services. AWS Whitepaper.
- **Literatura plotësuese / artikuj**
- Mora, H. et al. (2024). Mobile Cloud Computing Paradigm: A Survey of Architecture, Applications, and Challenges. International Journal of Communication Systems.
- Bashir, S. R. et al. (2024). A Narrative Review of Identity, Data and Location Privacy in Mobile Crowdsourcing and Edge Computing. Electronics, 13(21), 4228.
- Liu, J. et al. (2022). Energy-efficient allocation for multiple tasks in mobile edge computing. Journal of Cloud Computing, 11, Article 68.
- Amazon Web Services. (2024). AWS Well-Architected Framework

Ushtrimet



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AWS SimuLearn: Cloud Computing Essentials

AWS SimuLearn | ★ 4.7 (6741) | 1h | English [+6 more](#)

You're viewing this training as part of a learning plan: [AWS SimuLearn: Cloud Practitioner](#)



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Outline

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Cloud Computing Essentials

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BACKGROUND INFO

- Mag. Ing. Arben Lila
- Computer Science, Zagreb
- Communication Networks, Phd Cnd., Wien
- Ligjerues ne AAB
- Aktivist per mbrojtjen e shendetit dhe mjedisit
- Bjeshkatar, vrapues dhe çiklist

Njohuritë që do t'i fitoni

- Të dalloni skemat e ndryshme të qasjes në rrjet
- Të identifikoni fleksibilitetin e aplikacioneve Cloud
- Të dini llojet e shërbimeve Cloud
- Të krahasoni mekanizmat e sigurisë së informatave në Cloud
- Të aplikoni teknikat e mbrojtjes së privatësisë dhe besueshmërisë
- Të njiheni me trendet e reja në llogaritjen Cloud

Vlerësimi

Format e vlerësimit	Përqindja		Përqindja totale	Nota përfundimtare
Testi I	30		<50%	Nuk kalon
Testi II	30		>50%	Kalon
Projekti	30		51-60%	6
Pjesemarrja	10		61-70%	7
Provimi	100		71-80%	8
			81-90%	9
			91-100%	10

Projekti

Do të realizoni një Cloud projekt

Projekti vlerësohet me 30 pikë

Hyrje në llogaritjen Cloud & Infrastruktura e Cloud

What is Cloud Computing?

- What do you think?

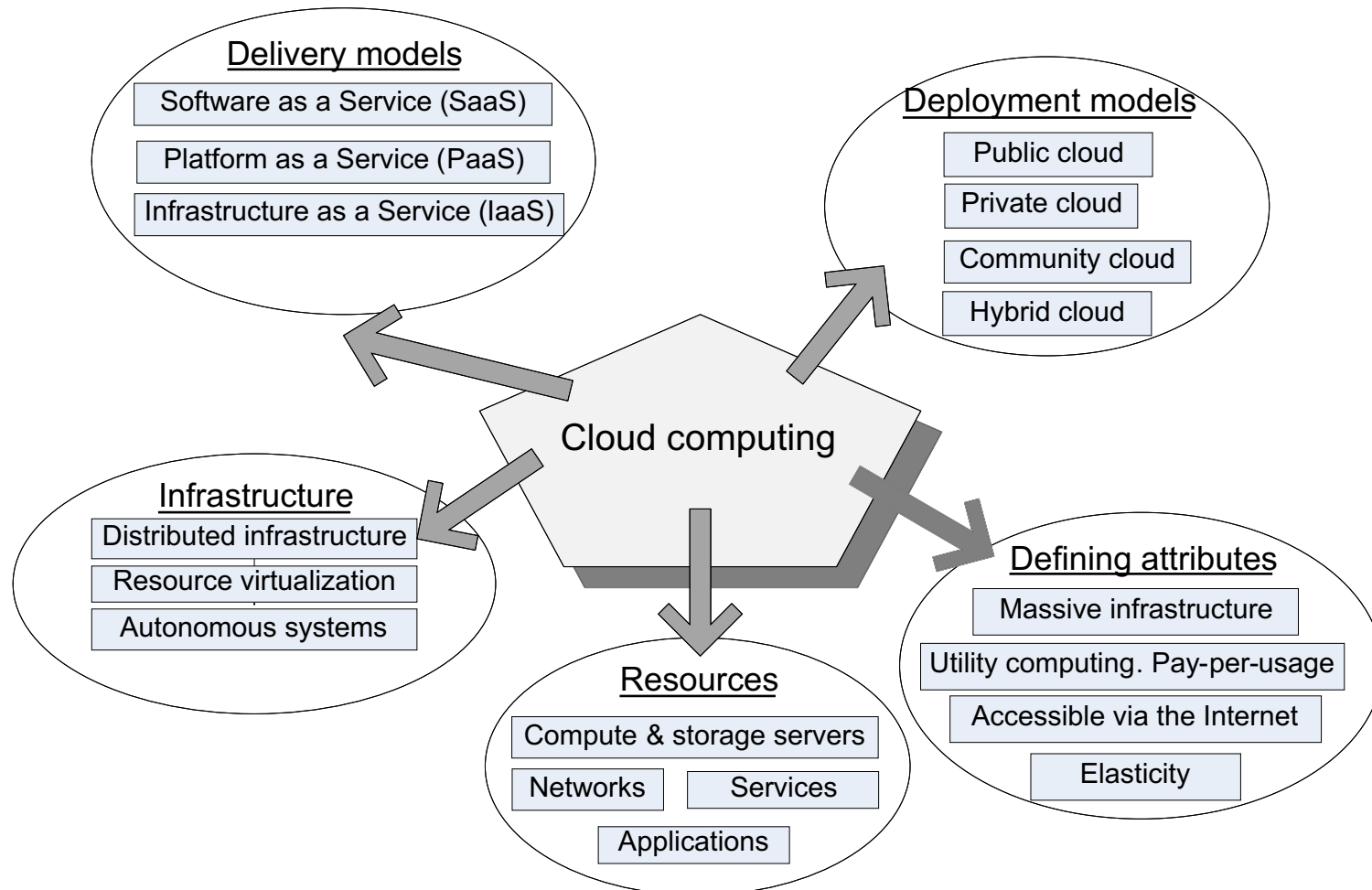
Cloud computing

- Is a technology that delivers computing services, including
 - servers,
 - storage,
 - databases,
 - networking,
 - software,
 - analytics,
 - and more, over the internet (cloud) rather than relying on local hardware.

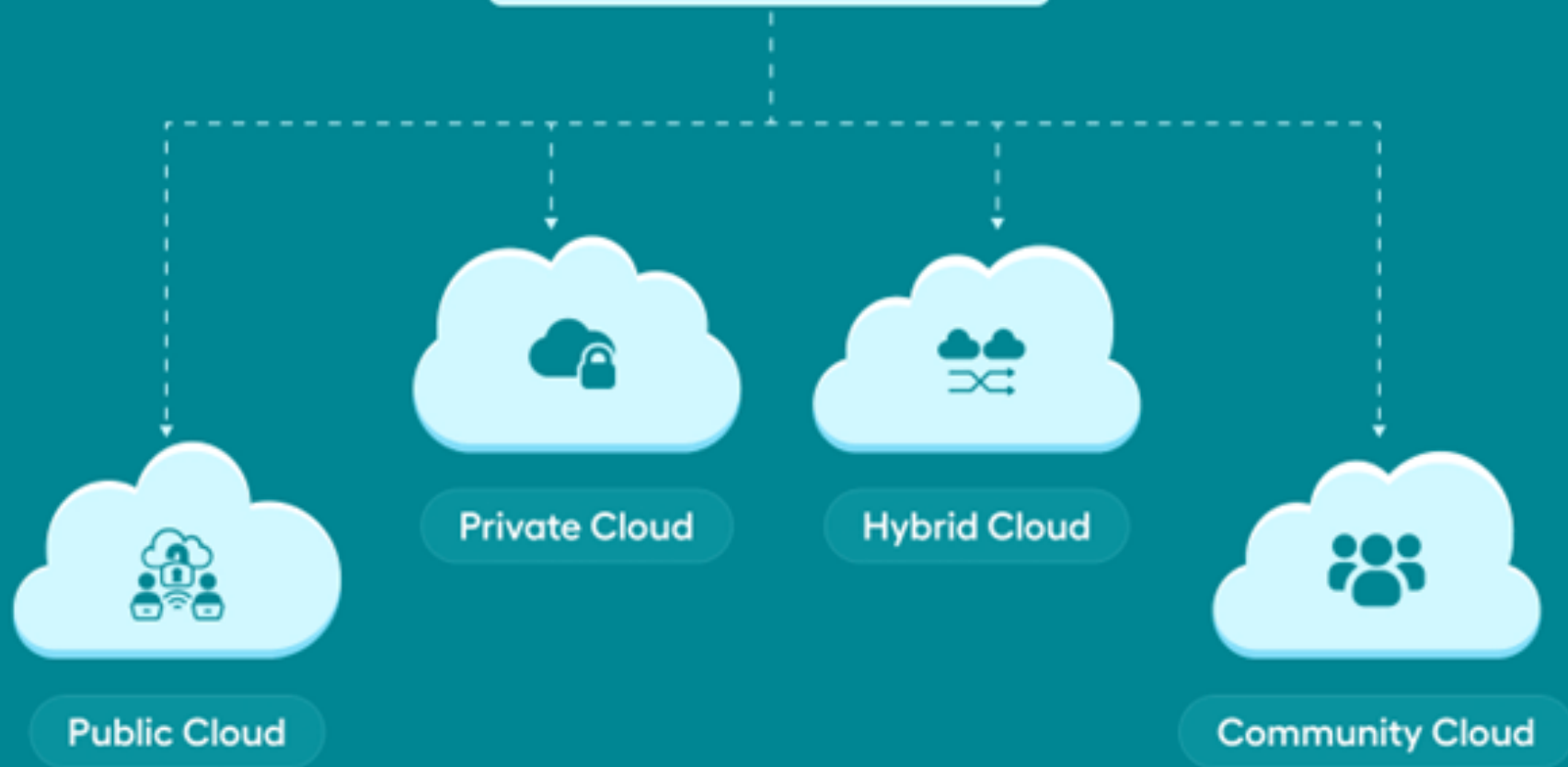
Fundamental Concepts

- Location Irrelevance: Users do not typically concern themselves with the physical location of the services or data centers because the cloud abstracts away those details
- History: The idea of computing as a service dates back to the 1960s, while the term "cloud computing" emerged in the early 2000s

Cloud Computing Models, Resources, Attributes

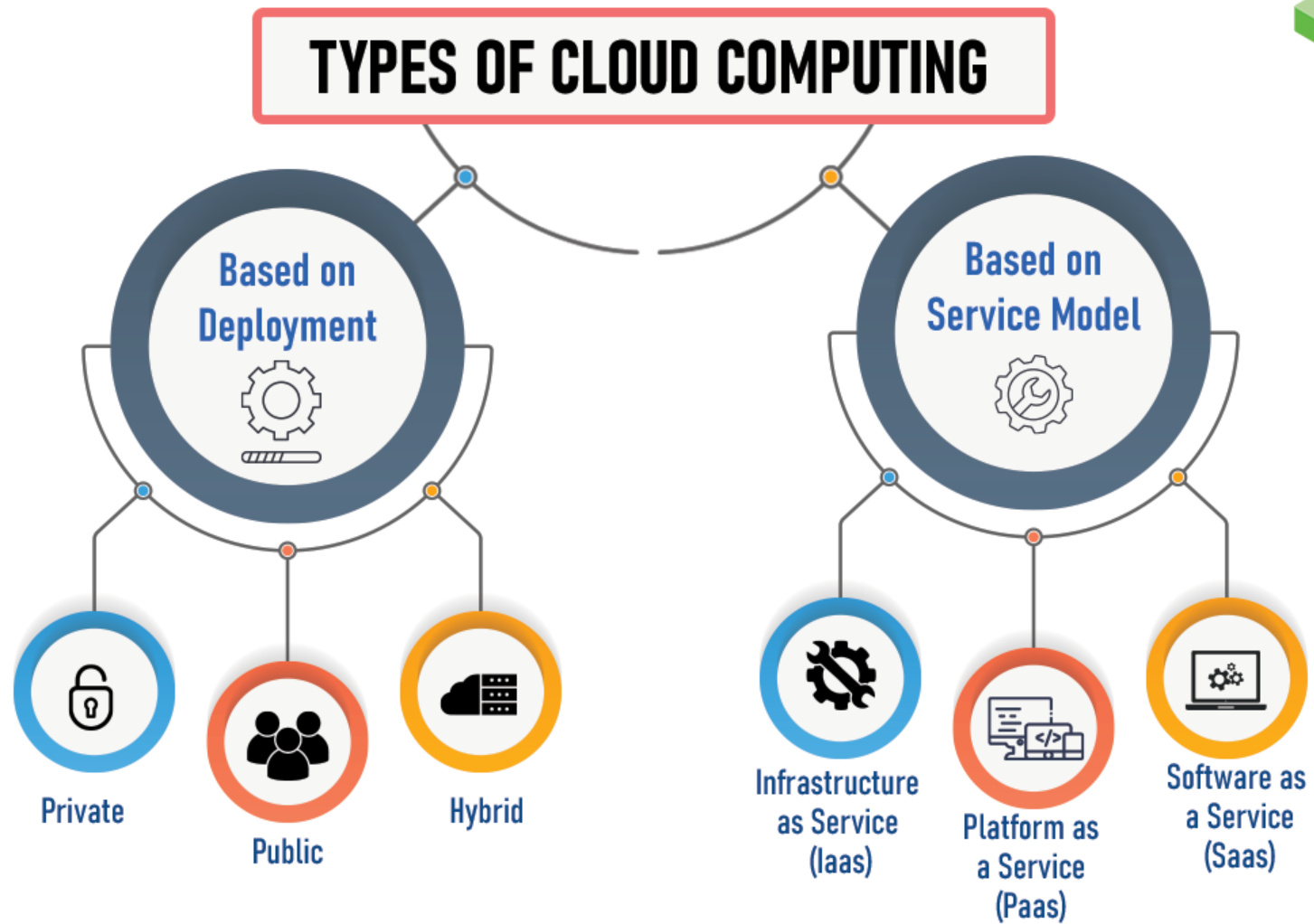


Types of Cloud



Types of Cloud Computing

- IaaS (Infrastructure as a Service): Providers supply compute, storage, and networking capabilities
- PaaS (Platform as a Service): Supplies a development environment for building and deploying applications
- SaaS (Software as a Service): Offers ready-made applications accessible via web browsers



Cloud Service Models

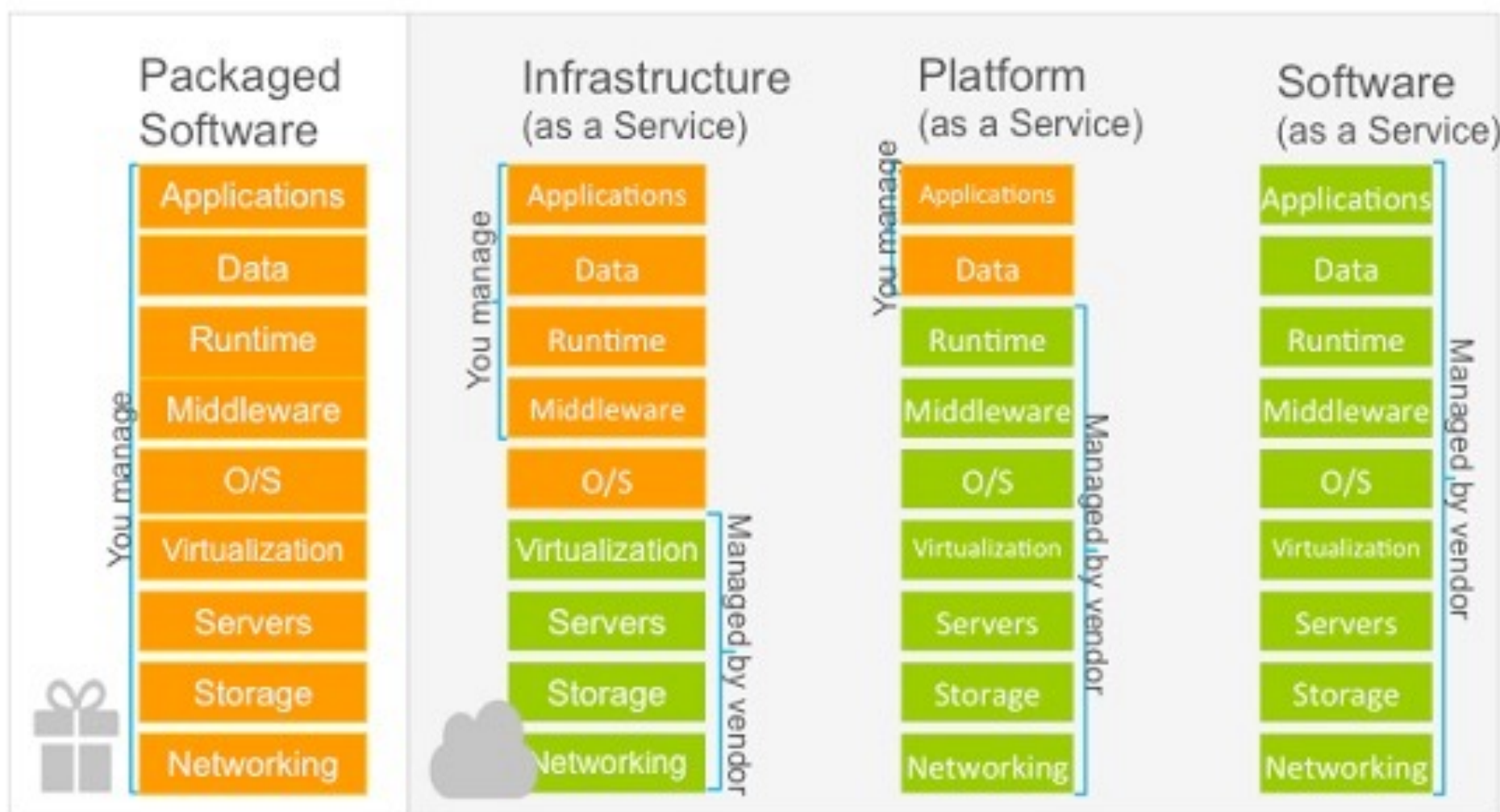
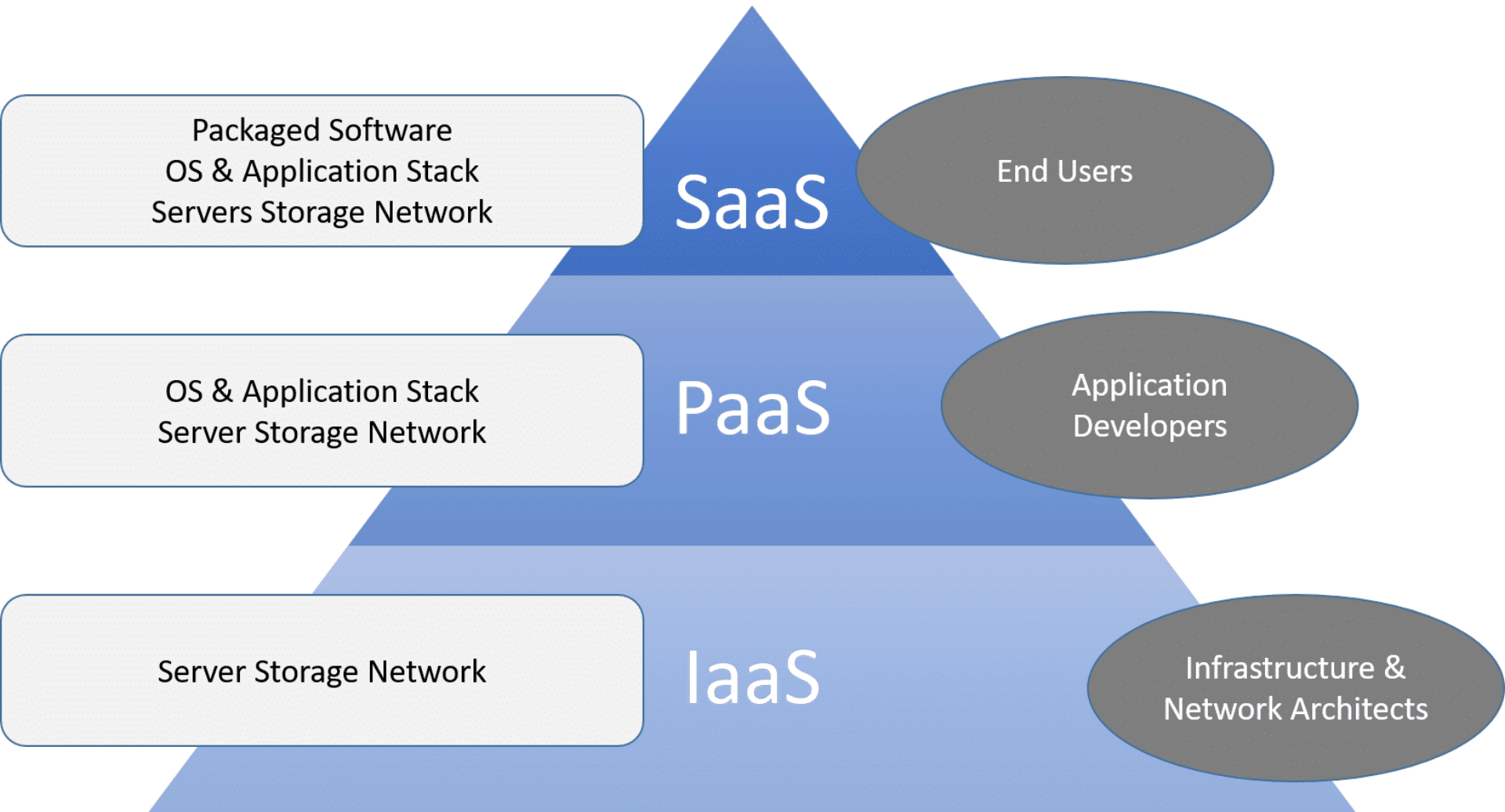


Figure 1.

Source: Microsoft Azure

Cloud Service Models



Cloud activities

- Service management and provisioning including:
 - Virtualization.
 - Service provisioning.
 - Call center.
 - Operations management.
 - Systems management.
 - QoS management.
 - Billing and accounting, asset management.
 - SLA management.
 - Technical support and backups.

Cloud activities (cont' d)

- Security & Privacy in Cloud Computing (Zero Trust, AI Security) including:
 - ID and authentication.
 - Certification and accreditation.
 - Intrusion prevention.
 - Intrusion detection.
 - Virus protection.
 - Cryptography.
 - Physical security, incident response.
 - Access control, audit and trails, and firewalls.

Cloud activities (cont' d)

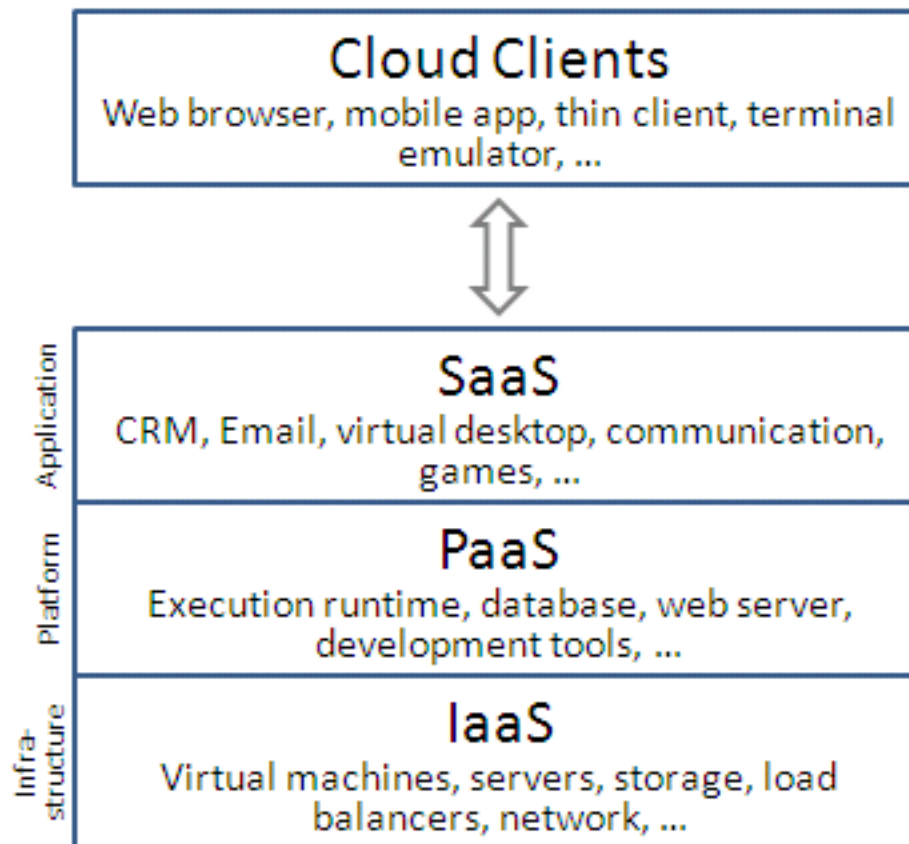
- Customer services such as:
 - Customer assistance and on-line help.
 - Subscriptions.
 - Business intelligence.
 - Reporting.
 - Customer preferences.
 - Personalization.
- Integration services including:
 - Data management.
 - Development.

Ethical issues

- Paradigm shift with implications on computing ethics:
 - The control is relinquished to third party services.
 - Data is stored on multiple sites administered by several organizations.
 - Multiple services interoperate across the network.
- Implications:
 - Unauthorised access.
 - Data corruption.
 - Infrastructure failure, and service unavailability.

Cloud Delivery Models

1. **Software as a Service (SaaS)** (high level)
2. **Platform as a Service (PaaS)**
3. **Infrastructure as a Service (IaaS)** (low level)



Advantages?

- Cost Savings: Organizations no longer need to purchase and maintain expensive hardware and software, reducing capital expenditures
- Scalability: Resources can easily be scaled up or down based on demand
- Accessibility: Data and applications can be accessed from anywhere with an internet connection
- Security: Cloud providers implement robust encryption and security measures to protect customer data

Challenges

- Data Privacy and Security Risks: Moving sensitive data offsite raises concerns about confidentiality and protection against cyber threats
- Vendor Lock-in: Choosing specific cloud providers might limit flexibility and increase dependence

Leading Cloud Providers

1. Amazon Web Services (AWS):

- Offering a vast range of services, AWS is a leading and widely adopted cloud platform, providing computing power, storage, and databases, among other services.

2. Microsoft Azure:

- Microsoft's cloud platform, Azure, provides a comprehensive set of services, including virtual machines, databases, AI, and analytics tools.

3. Google Cloud Platform (GCP):

- Google Cloud offers a suite of cloud services, including computing, storage, machine learning, and data analytics, leveraging Google's infrastructure.

4. IBM Cloud:

- IBM Cloud provides a diverse set of cloud computing services, including infrastructure as a service (IaaS), platform as a service (PaaS), and software as a service (SaaS).

5. Alibaba Cloud:

- As a major cloud provider in Asia and globally, Alibaba Cloud offers a wide range of cloud services, including computing, data storage, and AI.

Leading Cloud Providers

6. Oracle Cloud:

1. Oracle Cloud provides cloud infrastructure and a suite of enterprise cloud services, emphasizing its strengths in databases and enterprise applications.

7. VMware Cloud:

1. Known for its virtualization solutions, VMware extends its services to the cloud, offering a hybrid cloud platform for businesses.

8. Cisco Webex Teams:

1. Cisco offers cloud-based collaboration services, including video conferencing, messaging, and file sharing, under the Webex Teams platform.

9. Red Hat OpenShift:

1. Red Hat's OpenShift is a Kubernetes-based container platform that provides a consistent environment for deploying and managing applications across various infrastructures.

10. DigitalOcean:

1. DigitalOcean is known for its simplicity and developer-friendly approach, offering cloud computing solutions, including scalable virtual servers (Droplets) and managed databases.

Latest Trends in Cloud & Mobile Cloud Computing

- 1.AI-as-a-Service:** Cloud computing infrastructure will play a significant role in making AI accessible to businesses, enabling them to leverage its economic and social benefits
- 2.Hybrid and Multi-Cloud Strategies:** The number of large organizations with a multi-cloud strategy is expected to rise, offering cost and flexibility advantages while adding complexity to data governance and integration with legacy systems
- 3.Real-Time Cloud Infrastructure:** Organizations will increasingly look to leverage real-time data for up-to-the-minute insights and improved efficiency

Latest Trends in Cloud & Mobile

Cloud Computing

4. Cloud-Driven Innovation and Transformation:

Adopting cloud computing technology can serve as a gateway to emerging technologies like IoT, blockchain, and quantum computing

5. Cloud Security and Resilience:

Encryption, authentication, and disaster recovery will be in high demand as businesses prioritize security and resilience in their cloud infrastructure

6. Sustainable Cloud Computing:

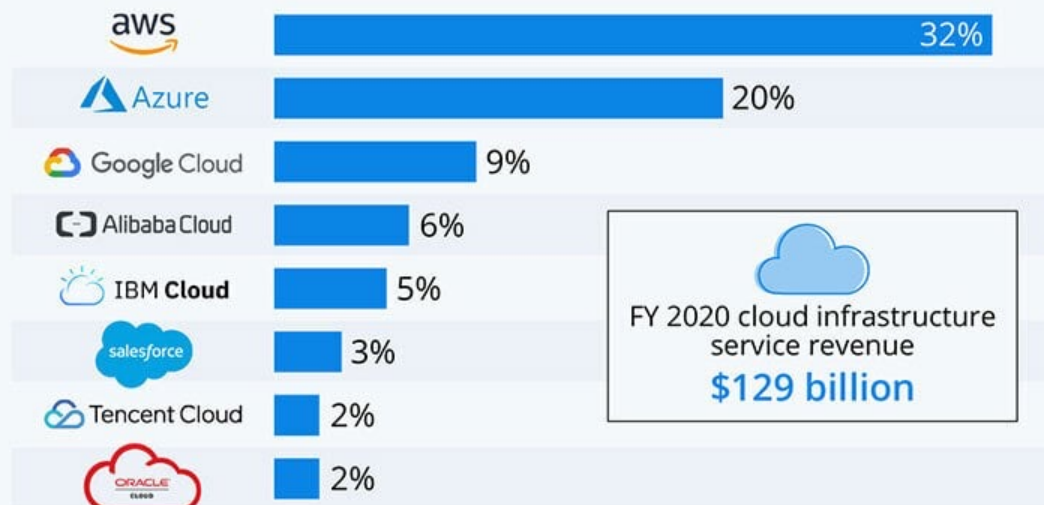
Cloud providers are making net-zero commitments and focusing on reducing the environmental impact of their services

Latest Trends in Cloud & Mobile Cloud Computing

7. **Simplified Cloud Computing:** Low-code/no-code tools and drag-and-drop interfaces are democratizing the use of cloud services and tools
8. **Privacy in the Cloud:** As businesses pass data to third-party cloud providers, managing privacy implications will remain an important theme in cloud computing
9. **Serverless and Pay-As-You-Go Cloud:** The serverless model eliminates the need for businesses to manage their own servers, allowing them to pay only for the resources they directly use
10. **Continued Investment in Data:** The cloud will continue to support resources such as scalable storage, servers, and databases, enabling businesses to analyze, store, and collate data

Amazon Leads \$130-Billion Cloud Market

Worldwide market share of leading cloud infrastructure service providers in Q4 2020*



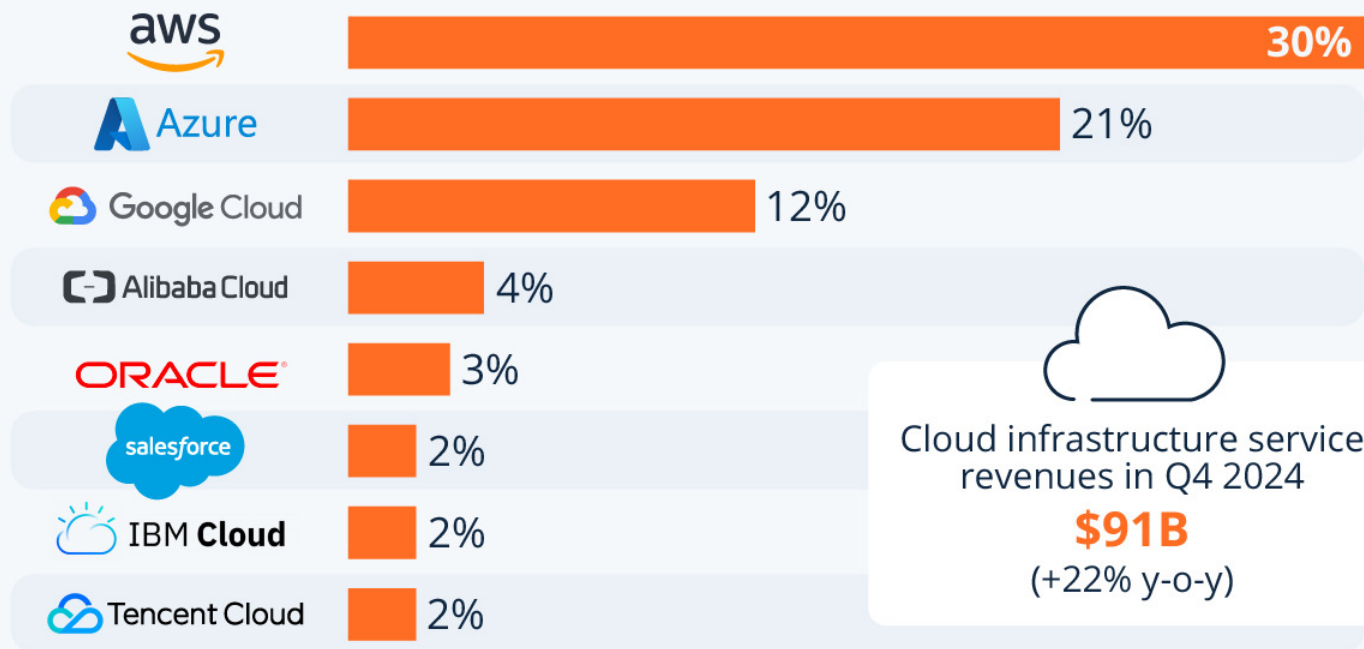
* includes platform as a service (PaaS) and infrastructure as a service (IaaS) as well as hosted private cloud services

Source: Synergy Research Group



Amazon and Microsoft Stay Ahead in Global Cloud Market

Worldwide market share of leading cloud infrastructure service providers in Q4 2024*



* Includes platform as a service (PaaS) and infrastructure as a service (IaaS) as well as hosted private cloud services

Source: Synergy Research Group



Dashboard – Wordpress Map

console.cloud.google.com/home/dashboard?inv=1&inv=Abr3Jg&project=wordpress-map-407911

Better World Quote... Refocusing school s... FJALORI I GJUHËS S... www.assembly-koso... Wegolo - Sucherge... Enable Visual Builder

Google Cloud

Wordpress Map

Search (/) for resources, docs, product... Search

1

Cloud overview

Solutions

PINNED PRODUCTS

APIs & Services

Billing

IAM & Admin

Marketplace

Compute Engine

Kubernetes Engine

Cloud Storage

BigQuery

VPC Network

Cloud Run

DASHBOARD

ACTIVITY

RECOMMENDATIONS

CUSTOMIZE

Project info

Project name
Wordpress Map

Project number
191114985598

Project ID
wordpress-map-407911

ADD PEOPLE TO THIS PROJECT

Go to project settings

APIs

Requests (requests/sec)

0.012/s

0.010/s

0.008/s

0.006/s

0.004/s

8:30 9 PM

Requests: 0.011/s

Go to APIs overview

Google Cloud Platform status

All services normal

Go to Cloud status dashboard

Billing

Estimated charges USD \$0.00

For the billing period Mar 1 – 12, 2025

Take a tour of billing

View detailed charges

Monitoring

Create my dashboard

Set up alerting policies

Resources

BigQuery
Data warehouse/analytics

SQL
Managed MySQL, PostgreSQL, SQL Server

Upcoming Earnings

Search

ENG

9:27 PM 3/12/2025

A Tour of Google Cloud Hands-on Labs

 Lab  45 minutes  No cost  Introductory

★★★★★ [Rate Lab](#)

 This lab may incorporate AI tools to support your learning.

Lab instructions and tasks



GSP282

Overview

Lab fundamentals

Task 1. Access the Cloud console

Task 2. View projects in the Cloud console

Task 3. Review and modify roles and permissions

Task 4. Enable APIs and services

Task 5. End your lab